Naveen Jindal School of Management

Business Analytics and Artificial Intelligence (BS)

Bachelor of Science in Business Analytics and Artificial Intelligence

Degree Requirements (120 semester credit hours)

View an Example of Degree Requirements by Semester

Faculty

FACG> jsom-business-analytics-bs

Professors: Ashiq Ali, Alain Bensoussan, Gary Bolton, Metín Çakanyildirim, Huseyin Cavusoglu, Jianqing Chen, William M. Cready, Gregory G. Dess, Umit G. Gurun, Dorothée Honhon, Kyle Hyndman, Varghese S. Jacob, Sanjay Jain, Ganesh Janakiraman, Elena Katok, Dmitri Kuksov, Nanda Kumar, Seung-Hyun Lee, Zhiang (John) Lin, Sumit K. Majumdar, Stanimir Markov, Amit Mehra, Syam Menon, Vijay S. Mookerjee, B. P. S. Murthi, Vikram Nanda, Özalp Özer, Mike W. Peng, Hasan Pirkul, Cuili Qian, Suresh Radhakrishnan, Srinivasan Raghunathan, Ram C. Rao, Brian Ratchford, Michael J. Rebello, Gil Sadka, Sumit Sarkar, Suresh P. Sethi, Kathryn E. Stecke, Riki Takeuchi, Wing Kwong (Eric) Tsang, Jun Xia, Ying Xie, Harold Zhang, Zhiqiang (Eric) Zheng

Associate Professors: Mehmet Ayvaci, Nina Baranchuk, Zhonglan Dai, Rebecca Files, Michael Hasler, Bin Hu, Surya N. Janakiraman, Robert L. Kieschnick Jr., Atanu Lahiri, Jun Li, Ningzhong Li, Maria Loumioti, Lívia Markóczy, Ramachandran (Ram) Natarajan, Naim Bugra Ozel, H. Dennis Park, Anyan Qi, Young U. Ryu, Serdar Simsek, Harpreet Singh, Upender Subramanian, Shouqiang Wang, Kelsey D. Wei, Han (Victor) Xia, Yexiao Xu, Alejandro Zentner, Jieying Zhang, Zhe (James) Zhang, Feng Zhao, Yibin Zhou

Assistant Professors: Khai Chiong, Rafael Copat, Soraya Fatehi, Andrew Frazelle, Ying Huang, Joonhwi Joo, Sora Jun, Jason Kautz, Tongil Kim, Sheen Levine, Christopher Mace, Samir Mamadehussene, Jean-Marie Meier, Radha Mookerjee, Jedson Pinto, Ignacio Rios Uribe, Alejandro Rivera Mesias, Simon Siegenthaler, Kirti Sinha, Xiaoxiao Tang, Shervin Tehrani, Ashwin Venkataraman, Guihua Wang, Hongchang Wang, Pingle Wang, Junfeng Wu, Steven Xiao, Yingjie Zhang

Associate Professors Emeriti: J. Richard Harrison, Jane Salk, David J. Springate

Clinical Professors: John Barden, Britt Berrett, Abhijit Biswas, Shawn Carraher, Larry Chasteen, Paul Convery, Howard Dover, Randall S. Guttery, William Hefley, Robert Hicks, Marilyn Kaplan, Sonia Leach, Peter Lewin, Jeffrey Manzi, John McCracken, Divakar Rajamani, Daniel Rajaratnam, Kannan Ramanathan, Mark Thouin, McClain Watson, Jeff Weekley, Habte Woldu, Fang Wu

Clinical Associate Professors: Shawn Alborz, Dawn Owens, Carolyn Reichert, Avanti P. Sethi, Ramesh Subramoniam, Aysegul Toptal, David Widdifield

Clinical Assistant Professors: Athena Alimirzaei, Christina (Krysta) Betanzos, Moran Blueshtein, Jerome Gafford, Jeffery (Jeff) Hicks, Revansiddha Khanapure, Liping Ma, Ravi Narayan, Parneet Pahwa, Jason Parker, Nassim Sohaee

Professors of Instruction: Semiramis Amirpour, Mary Beth Goodrich, Chris Linsteadt, Matt Polze, Luell (Lou) Thompson

Associate Professors of Instruction: Vivek Arora, Judd Bradbury, Monica E. Brussolo, Amal El-Ashmawi, Ayfer Gurun, Maria Hasenhuttl, Julie Haworth, Thomas (Tom) Henderson, Kathryn Lookadoo, Sarah Moore, Hubert Zydorek

Assistant Professors of Instruction: Negin Enayaty Ahangar, Joseph Mauriello, Rasoul Ramezani, Gaurav Shekhar

Professors of Practice: Gregory Ballew, Tiffany A. Bortz, Ranavir Bose, Alexander Edsel, Rajiv Shah

Associate Professors of Practice: Richard Bowen, Jackie Kimzey, David Parks, Margaret Smallwood, Steven Solcher, Kathy Zolton

Assistant Professors of Practice: Khatereh Ahadi, Steven Haynes, Abu Naser Islam, Edward Meda, Paul Nichols, Timothy Stephens, Guido Tirone, Robert Wright

Students must meet all course prerequisites.

I. Core Curriculum Requirements: 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

Communication: 6 semester credit hours

Select any 6 semester credit hours from Communication Core courses

Mathematics: 3 semester credit hours

MATH 1325 Applied Calculus I[1, 2]

Or select any 3 semester credit hours from Mathematics Core courses

Life and Physical Sciences: 6 semester credit hours

Select any 6 semester credit hours from Life and Physical Sciences Core courses

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from Language, Philosophy and Culture Core courses

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from Creative Arts Core courses

American History: 6 semester credit hours

Select any 6 semester credit hours from American History Core courses

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from Government/Political Science Core courses

Social and Behavioral Sciences: 3 semester credit hours

Choose one of the following:

BA 1310 Making Choices in Free Market Systems[1, 2]

or BA 1320 Business in a Global World[1, 2]

or ECON 2301 Principles of Macroeconomics[1, 2]

or ECON 2302 Principles of Microeconomics[1, 2]

Or select any 3 semester credit hours from Social and Behavioral Sciences Core courses

Component Area Option: 6 semester credit hours

Choose one of the following:

BA 1310 Making Choices in Free Market Systems[1, 2]

or BA 1320 Business in a Global World[1, 2]

or ECON 2301 Principles of Macroeconomics[1, 2]

or ECON 2302 Principles of Microeconomics[1, 2]

And select 3 semester credit hours from Component Area Option Core

Or select any 6 semester credit hours from Component Area Option Core courses

II. Major Requirements: 76 semester credit hours

Major Preparatory Courses: 18 semester credit hours beyond Core Curriculum

ACCT 2301 Introductory Financial Accounting

ACCT 2302 Introductory Management Accounting[1]

BLAW 2301 Business and Public Law[1]

ITSS 3311 Introduction to Programming[3]

OPRE 3333 Quantitative Business Analysis[1, 4]

OPRE 3360 Managerial Methods in Decision Making Under Uncertainty[4]

Complete the following courses if not already completed as part of the Core Curriculum Requirements:

MATH 1325 Applied Calculus I[1, 2] (Required for all JSOM undergraduate students. May elect to substitute MATH 2413 or MATH 2417 for MATH 1325.)

Choose two of the following if not completed in Core Curriculum:

BA 1310 Making Choices in Free Market Systems[1, 2]

BA 1320 Business in a Global World[1, 2]

ECON 2301 Principles of Macroeconomics[1, 2]

ECON 2302 Principles of Microeconomics[1, 2]

Major Core Courses: 28 semester credit hours

BA 1105 Professional Development[1] (JSOM first-time-in-college freshmen are required to take BA 1105 in their first semester)

or BA 3105 Professional Development (Transfer students and students new to JSOM are required to take BA 3105 in their first semester)

BCOM 3300 Business Communication

BUAN 3301 AI in Business

FIN 3320 Business Finance

IMS 3310 International Business

ITSS 3300 Information Technology for Business

MKT 3300 Principles of Marketing

OBHR 3310 Organizational Behavior[5]

or OBHR 3330 Human Resource Management in Business[5]

OPRE 3310 Supply Chain and Operations Management

Experiential Learning

A capstone course is required for all undergraduate JSOM students.

BUAN 4395 Capstone Senior Project - Business Analytics

or BPS 4395 Capstone Senior Project - Business

A practicum experience of at least 160 working hours is required, with registration in the course below.

BUAN 4090 Business Analytics Internship[6]

Students are required to complete 100 hours of community engagement.

BA 4095 Social Sector Engagement and Community Outreach Practicum[7]

Major Related Courses: 21 semester credit hours

BUAN 4320 Database Fundamentals for Analytics

BUAN 4351 Foundations of Business Intelligence

BUAN 4353 Business Analytics

BUAN 4355 Data Visualization

BUAN 4373 Data Science for Business Applications

BUAN 4381 Object Oriented Programming with Python

BUAN 4383 Machine Learning for Business Analytics

Guided Electives: 9 semester credit hours

The BS BUAN degree program gives students the opportunity to focus on a specific Concentration to obtain in-depth knowledge in a specific business area. Students will choose 9 semester credit hours from one of the following Concentrations:

A. Marketing Analytics and AI Concentration

Students who pursue this concentration learn skills necessary for using data to gather deeper consumer insights, optimizing marketing objectives, and in general, achieving a higher return on investment. Students completing this concentration may pursue careers as Marketing and Research Analysts, Social Media and Digital Marketing Specialists.

BUAN 4337 Marketing Analytics

MKT 4339 AI-Driven Content Creation and Campaign Management

And select one of the following:

MKT 4330 Digital and Internet Marketing

MKT 4331 Digital Prospecting

MKT 4334 Social Media Marketing

MKT 4336 E-Retailing

MKT 4341 Predictive Analytics

MKT 4390 Advanced Marketing Analytics

B. Operations and Supply Chain Analytics and AI Concentration

Students who pursue this concentration learn to apply analytical concepts in diverse areas in supply chain industry such as integrated supply chain management, project management and inventory management to name a few. Students completing this concentration may pursue careers as Supply Chain Analysts, Project Managers and Product Analysts.

OPRE 4320 Supply Chain Information Systems: SAP, AI, and Blockchain Innovations

OPRE 4330 Global Logistics and Inventory Management

And select one of the following:

OPRE 3320 Integrated Supply Chain Management

OPRE 4310 Lean and Six Sigma Processes

OPRE 4340 Purchasing and Sourcing Management

OPRE 4393 A.I. in Supply Chain Management

BUAN 4357 Supply Chain Analytics, AI, and Advanced Solutions

C. Finance and Risk Analytics Concentration

Students who choose this concentration will learn both the qualitative and quantitative tools necessary to enter one of the many different areas within the finance risk management and insurance industries. Students completing this concentration may pursue careers as Financial Analysts, Quantitative Risk Analysts and Credit Risk Model Developers.

FIN 4300 Investment Management

FIN 4338 Foundations of Risk Analytics and Applications

And select one of the following:

FIN 3390 Introduction to Financial Modeling

FIN 4322 FinTech Markets

FIN 4345 Financial Information and Analysis

FIN 4346 Applied Machine Learning in Finance

FIN 4352 Financial Risk Management

D. Information Technology Concentration

Students who choose this concentration will learn key fundamentals of how businesses design and use information technology to optimize their business processes. Combined with business analytics knowledge students completing this concentration may pursue careers as Business Intelligence Analysts, IT Analysts, Cybersecurity Analysts, and Systems Analysts.

ITSS 4330 Systems Analysis and Design

ITSS 4360 Network and Information Security

And select one of the following:

ITSS 4340 Enterprise Resource Planning

ITSS 4344 CRM using Salesforce

ITSS 4356 Data Governance

ITSS 4361 Information Technology Cybersecurity

ITSS 4362 Cybersecurity Governance

ITSS 4371 Cloud Computing with AWS

E. Data Science Concentration

Students who choose this concentration will advance their knowledge into domains of advanced applied machine learning, web analytics, artificial intelligence, and big data analytics. Students may pursue job titles such as Data Scientist, Data Mining Engineer, Data Analyst, Decision Scientist, Machine Learning Scientist, Data Manager and Data Architect.

BUAN 4322 Causal Analysis and A/B Testing

BUAN 4354 Advanced Big Data Analytics

And select one of the following:

BUAN 4321 Business Data Engineering

BUAN 4350 Spreadsheet Modeling and Analytics with AI

BUAN 4352 Introduction to Web Analytics

F. Human Resources (HR) Analytics Concentration

Students who pursue this concentration will be able to apply their analytical knowledge to analyze workforce data, improve employee performance, and optimize HR decision-making. They will gain expertise in leveraging data to address talent management, recruitment, and retention challenges. Students pursuing this concentration and any of the above, may pursue the job title of Business Analysts or HR Analysts.

OBHR 4331 Compensation and Benefits Administration

OBHR 4337 HR Analytics

OBHR 4339 People Technology Leadership and Artificial Intelligence

G. Applied Artificial Intelligence Concentration

Students pursuing the Applied AI concentration will learn how to design and implement AI-driven solutions for real-world business challenges, leveraging machine learning, natural language processing, and generative AI tools. They will also develop a strong understanding of the ethical, responsible, and strategic use of AI in business decision-making and innovation.

BUAN 4321 Business Data Engineering

BUAN 4323 Applied Generative AI for Business

BUAN 4358 Responsible AI Practices in Business

H. General Concentration

Students who pursue this concentration have the flexibility to apply their analytical knowledge and skills to diverse areas not limited to business domains of Marketing, Finance, Supply Chain, or IT, but also in areas of Litigation, Social Sciences, Sports Analytics and many more. Students pursuing this concentration and any of the above, may pursue the job title of Business Analysts.

Any combination of approved upper-level guided elective courses listed in concentrations A, B, C, D, E, or F may be used to satisfy the required 9 hours of guided elective credit.

Students in the Jindal Undergraduate Research Scholars program should take BA 3340, BA 3350, and up to 3 SCH of an Individual Study based on the major to satisfy JURS program requirements. These credits may be applied to the general concentration.

III. Elective Requirements: 2 semester credit hours

Free Electives: 2 semester credit hours

Both lower- and upper-division courses may count as electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

1. Indicates a prerequisite class to be completed before enrolling for upper-division classes.

2. A required Major course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.

3. Completion of CS 1436 and/or CS 1337 with a C or better will satisfy ITSS 3311 with an additional 3 semester credit hours of ITSS upper-division courses (excluding ITSS 4301 and ITSS 4V90).

4. Students may use ENGR 2300, TE 2305, MATH 2418, or MATH 2333 as a substitute for OPRE 3333. Students may use CS 3341, ENGR, 3341, SE 3341, STAT 3341, STAT 2332, STAT 3355, STAT 3360, or STAT 4351 as a substitute for OPRE 3360.

5. Students who wish to pursue the HR Analytics concentration should take OBHR 3330 to comply with the pre-requisite requirements for courses listed in the concentration.

6. Students may fulfill the internship requirement with BUAN 4090 or BUAN 4V90 (1-3 semester credit hours). The zero-semester credit hour course BUAN 4090 is recommended as the most efficient way to satisfy this requirement. If the variable credit internship option is taken, semester credit hours apply as free elective credit towards degree requirements.

7. Students may fulfill the community engagement requirement with BA 4095 or ENTP 4340. The zero semester credit hour course BA 4095 is recommended as the most efficient way to satisfy this

requirement. If the variable credit community engagement option is taken, semester credit hours apply as free elective credit towards degree requirements.

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